

Lesson 1.9-1.10: Rational Functions Vertical Asymptotes and Holes

Topic 1.9: Rational Functions and Vertical Asymptotes

Topic 1.10: Rational Functions and Holes

In a previous lesson we defined a rational function as the ratio of two polynomials. The domain for polynomial functions is all real numbers, but the domain of a rational function will be restricted from any values of x that cause **division by zero**. Now, we turn our attention to defining vertical asymptotes and finding holes.



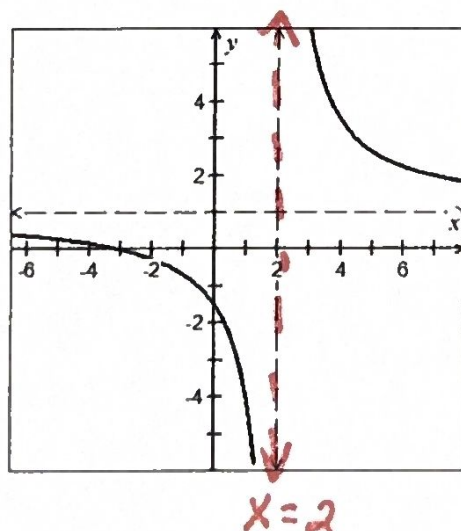
Characteristics of Rational Functions

Vertical asymptotes and holes in the graph of a rational function both occur when the denominator values equal zero.

- ❑ A **vertical asymptote** occurs when a factor in the denominator does not "cancel out" with factors in the numerator. That is, $x = 0$.
- ❑ A **hole** will occur when a common factor in the denominator "cancels out" with factors in the numerator.
- ❑ **Note:** The degree (multiplicity) of common factors and their location in the rational function will be important to consider in this lesson.

Vertical Asymptotes

$$f(x) = \frac{(x+3)}{(x-2)} \quad \begin{array}{l} x = -3 \text{ Zero} \\ x = 2 \text{ VA} \end{array}$$



$f(x)$ has a vertical asymptote at $x = 2$

$$\lim_{x \rightarrow 2^-} f(x) = -\infty$$

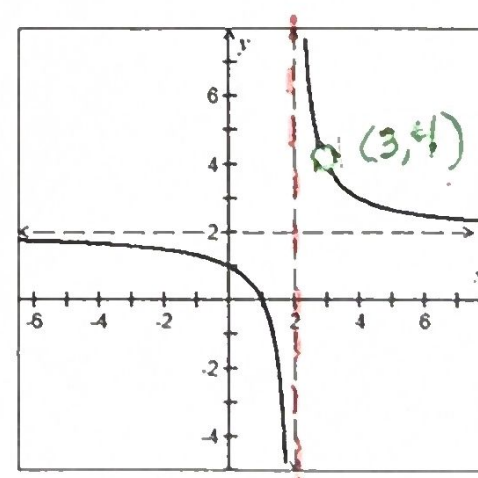
$$\lim_{x \rightarrow 2^+} f(x) = +\infty$$

from the
left

from the right

Holes $x = 3$ Hole

$$g(x) = \frac{2(x-3)(x-1)}{(x-3)(x-2)} \quad \begin{array}{l} x = 1 \text{ Zero} \\ x = 2 \text{ VA} \end{array}$$



$g(x)$ has a hole at $x = 3$ and a vertical asymptote at $x = 2$

$$\lim_{x \rightarrow 3^-} g(x) = 4$$

$$\lim_{x \rightarrow 3^+} g(x) = 4$$

Vertical Asymptotes and Multiplicities

Let's explore how we can use the repeated factors in the numerator and denominator to help identify where vertical asymptotes might be without having to simplify:

$$h(x) = \frac{\overset{\text{Hole}}{\cancel{(x-4)}^2}(x-5)}{(x-4)\cancel{(x-5)}^2} \quad \text{VA}$$

$$f(x) = \frac{\overset{\text{Hole}}{(x+2)}^2 \overset{\text{Hole}}{(x-3)} \overset{\text{Hole}}{(x+4)}^6}{(x+2)^2(x-3)(x+4)^2}$$

$$g(x) = \frac{\overset{\text{Zero}}{(x-6)} \overset{\text{Hole}}{\cancel{(x+3)}^2}}{\underbrace{x(x-1)}_{\text{VA}}(x+3)}$$

All three functions above have repeated factors in both the numerator and denominator. Without simplifying, how can you tell which factors will create a vertical asymptote?

Vertical asymptote: Bottom degree is larger and denominator does not cancel

Hole: Top degree is larger or top and bottom degrees are equal

EX #1: For each function below, determine the x-values for any vertical asymptotes or holes.

A. $f(x) = \frac{\cancel{(x-4)}(x+2)}{(x-3)\cancel{(x-4)}}$

Hole: $x = 4$

VA: $x = 3$

(Zero: $x = -2$)

B. $g(x) = \frac{\cancel{(x+3)}(x-1)^2}{\cancel{(x-1)}(x+3)^2}$

Hole: $x = 1$

VA: $x = -3$

(no zeros)

C. $h(x) = \frac{2}{x^3+5x}$
 $\quad \quad \quad x(x^2+5)$

No Holes

VA: $x = 0$

(no zeros)

D. $k(x) = \frac{x^2-9}{x^2+x-6}$ $\frac{\cancel{(x+3)}\cancel{(x-3)}}{\cancel{(x+3)}(x-2)}$

Hole: $x = -3$

VA: $x = 2$

(Zero: $x = 3$)



Limits and Vertical Asymptotes

We have used **limits** to describe the end behavior of both polynomial functions and rational functions. Now, we will use **limits** to describe the behavior of a function around a vertical asymptote. When looking at the graph of a rational function, the function will either increase without bound or decrease without bound, on either side of the asymptote. We will use **one-sided limits** to help determine the function's behavior on the both sides of the vertical asymptote.

Limit Notation	Verbally
Left Hand Limit: $\lim_{x \rightarrow a^-} f(x) = \pm\infty$	For input values near and less than a , the output values are increasing/decreasing without bound. "The limit as x approaches a from the <u>left</u> of $f(x)$ is positive/negative infinity."
Right Hand Limit: $\lim_{x \rightarrow a^+} f(x) = \pm\infty$	For input values near and greater than a , the output values are increasing/decreasing without bound. "The limit as x approaches a from the <u>right</u> of $f(x)$ is positive/negative infinity."

EX #2: Write a left and right limit statement as x approaches 1 for each of the following functions:

A. $f(x) = \frac{(x-5)(x+2)}{(x-1)}$ **VA: $x=1$**

$f(.9) = 118.9 \rightarrow \infty$
 $f(1.1) = -120.9 \rightarrow -\infty$

Left-hand limit: $\lim_{x \rightarrow 1^-} f(x) = \infty$

Right-hand limit: $\lim_{x \rightarrow 1^+} f(x) = -\infty$

B. $g(x) = \frac{(x+3)(x-1)}{(x+1)(x-1)}$ **Hole: $x=1$**

$g(1) = \frac{1+3}{1+1} = 2$ (or you could use .9 and 1.1 too)

Left-hand limit: $\lim_{x \rightarrow 1^-} g(x) = 2$

Right-hand limit: $\lim_{x \rightarrow 1^+} g(x) = 2$

EX #3: A rational function $r(x)$ has the following characteristics: **bottom**

Top Zeros at $x=4$ and $x=0$; vertical asymptotes at $x=-8$ and $x=-3$, a zero of $x=-3$ in both the numerator and denominator, and $\lim_{x \rightarrow -\infty} r(x) = -2$. Create an equation for $r(x)$.

Note: There may be multiple answers.

$$r(x) = \frac{-2x(x-4)(x+3)}{(x+8)(x+3)^2}$$

HA so LC must be -2
 and degrees must be equal

EX #4: Write an equation of a rational function with the following limit properties.

A. $f(2)$ is undefined

$$\lim_{x \rightarrow 2^-} f(x) = 3 \quad \lim_{x \rightarrow 2^+} f(x) = 3 \quad \text{Hole}$$

$$\lim_{x \rightarrow -4^-} f(x) = -\infty \quad \lim_{x \rightarrow -4^+} f(x) = +\infty \quad \text{VA}$$

$$f(x) = \frac{(x-2)}{(x+4)(x-2)}$$

B.

$g(-1)$ is undefined

$$\lim_{x \rightarrow -1^-} g(x) = 2 \quad \lim_{x \rightarrow -1^+} g(x) = 2 \quad \text{Hole}$$

$$\lim_{x \rightarrow 3^-} g(x) = +\infty \quad \lim_{x \rightarrow 3^+} g(x) = +\infty \quad \text{VA}$$

$$g(x) = \frac{x+1}{(x+1)(x-3)^2}$$

(Answers may vary)

because left and right end behavior are both $+\infty$

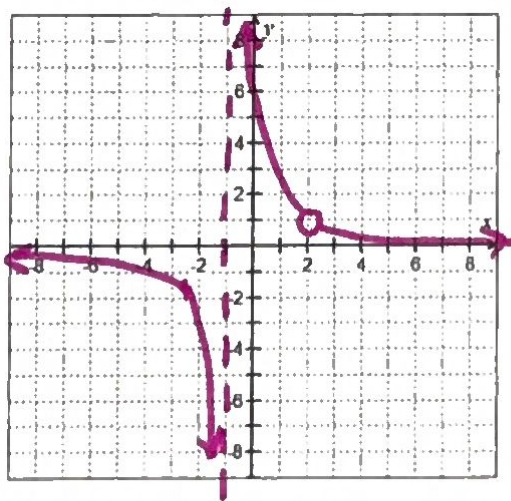
EX #5: Sketch a picture of a rational function that has the following properties.

A. $h(x)$ has a hole at $x = 2$

As x approaches -1 from the left, $h(x)$ decreases without bound. $-\infty$

As x approaches -1 from the right, $h(x)$ increases without bound. $+\infty$

VA @ $x = -1$



(graphs will vary)

B. $k(x)$ has holes at $x = -1$ and $x = 4$

As x approaches 2 from the left, $k(x)$ increases without bound. $+\infty$

As x approaches 2 from the right, $k(x)$ increases without bound. $+\infty$

VA @ $x = 2$

